

Project: Model Evaluation, Explainability, and Fairness

Insurance Claim Approval: Predicting Claim Outcomes

Ebubechi Amanda Enyi-Igbokwe

1. Business Overview

Insurance underwriting organizations process thousands of claims each year. Every approval represents a financial commitment, money paid out to a claimant and every denial represents a service decision that may be challenged through appeals, complaints, or regulatory escalation. When organisations rely on manual review alone, claims teams face inconsistent decisions, slow processing times, and uneven exposure to fraud and over-approval risk.

This assignment applies a Decision Tree classification model to a dataset of 380 insurance claims. The goal is to predict, at the point of submission, whether a specific claim is likely to be approved or denied. By scoring incoming claims in advance, claims teams can prioritise reviewer time, flag higher-risk submissions for additional scrutiny, and apply more consistent approval criteria across customers and regions.

The analysis goes beyond simple accuracy reporting. It evaluates the model using multiple classification metrics, explains predictions using SHAP and LIME, and assesses whether the model performs equitably across demographic and geographic subgroups. The report concludes with a business-oriented recommendation on model readiness.

2. Data Preparation

2.1. Dataset Overview

The dataset contains 380 records representing individual insurance claims. Each record has 19 columns spanning four categories:

- **Customer profile:** age, age_group, gender, region, customer_tenure_years, customer_segment

- **Policy and premium:** policy_type, annual_premium, coverage_amount, payment_history_score
- **Vehicle and driver:** vehicle_age_years, driving_experience_years
- **Claim details:** previous_claims, claim_amount, accident_severity, police_report_filed, claim_processing_days

The target variable is claim_approved (1 = approved, 0 = not approved). The class distribution is mildly imbalanced: 209 claims were approved (55.0%) and 171 were not (45.0%).

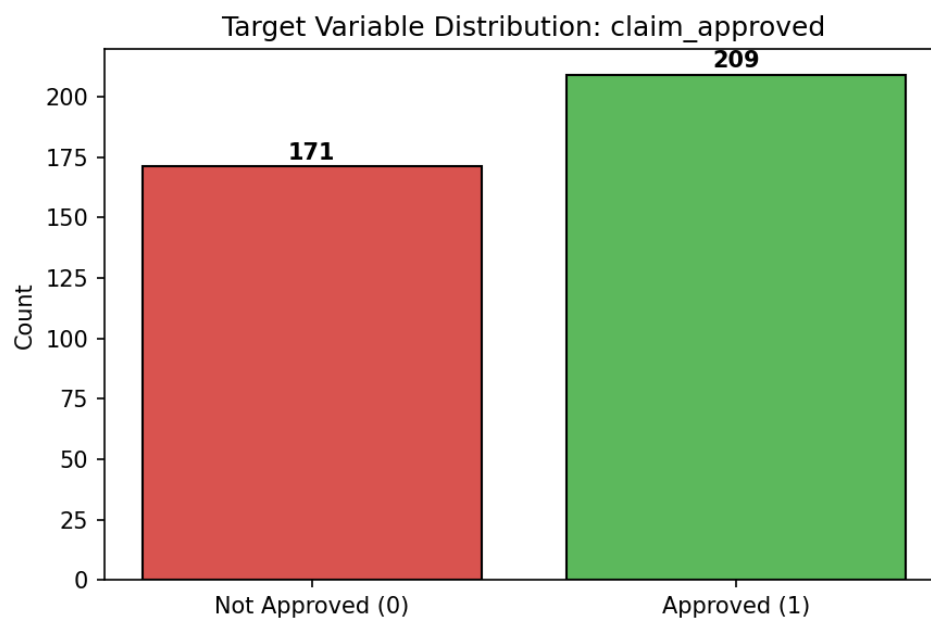


Figure 1: Target variable distribution 209 approved (55.0%), 171 not approved (45.0%)

2.2. Feature Design

Three columns were excluded from model input and retained only for fairness analysis: gender, age_group, and region. These are protected or group-sensitive attributes that should not drive predictions but must be monitored to assess model equity. The claim_id column was also dropped as a non-predictive unique identifier.

The remaining 14 columns formed the model input features, split into:

- **Numerical:** age, vehicle_age_years, driving_experience_years, annual_premium, coverage_amount, previous_claims, claim_amount, customer_tenure_years, payment_history_score, claim_processing_days
- **Categorical:** policy_type, customer_segment, accident_severity, police_report_filed

2.3. *Preprocessing Steps*

All preprocessing was applied using a scikit-learn Pipeline and ColumnTransformer to prevent data leakage:

- **Numerical features:** missing values imputed using the column median (SimpleImputer). Four columns had missing values: annual_premium (6 missing), claim_amount (7 missing), payment_history_score (7 missing), and police_report_filed (7 missing).
- **Categorical features:** missing values imputed with the most frequent value, then one-hot encoded (OneHotEncoder).
- The full preprocessing pipeline was fitted only on training data and applied to the test set, ensuring no information from the test set influenced preprocessing decisions.

The final processed feature matrix contained 21 columns after one-hot encoding expanded the categorical variables. No duplicate records were found in the dataset.

3. **Train/Test Split**

The dataset was split 80/20 into training and testing sets using stratified sampling to preserve the class ratio in both sets.

- **Training set:** 304 records (80%)
- **Testing set:** 76 records (20%)
- **Stratified split:** maintains the 55%/45% class ratio in both sets
- **random_state=42** used for reproducibility

The training set is used to teach the model the relationship between features and the target label. The testing set is held out completely and used only after training to evaluate how the model performs on data it has never seen. This separation is essential to obtain an honest estimate of real-world performance.

4. **Decision Tree Model**

4.1. *How Decision Trees Make Predictions*

A Decision Tree is a supervised machine learning algorithm that learns a series of yes/no decision rules from the training data. At each node in the tree, the algorithm selects the

feature and threshold that best separates the training records by their target label. This process repeats recursively until a stopping condition is reached, in this case, a maximum depth limit.

The primary advantage of Decision Trees is interpretability: a prediction can be traced step by step through the tree to understand exactly which conditions led to a particular output. This is especially valuable in regulated business contexts like insurance, where stakeholders need to explain and validate model decisions to claimants, auditors, and regulators.

4.2. Model Training

Two models were trained:

- **Unconstrained tree:** no depth limit, trained for comparison only. Reached depth 10, achieving high training accuracy through overfitting.
- **Controlled tree (max_depth=4):** selected as the final model. Restricts splits to 4 levels, preventing overfitting and improving generalisation to unseen data.

All evaluations, metrics, feature importances, SHAP values, and LIME explanations in this report use the controlled model (max_depth=4). The table below shows the first 15 test records with their actual labels, predicted labels, and whether the prediction was correct.

Record	Actual	Predicted	Correct	Interpretation
1	1	1	Yes	Correctly predicted approval
2	0	0	Yes	Correctly predicted denial
3	1	1	Yes	Correctly predicted approval
4	0	0	Yes	Correctly predicted denial
5	0	1	No	False Positive (predicted approval, was denied)
6	0	0	Yes	Correctly predicted denial
7	1	1	Yes	Correctly predicted approval
8	1	0	No	False Negative (predicted denial, was approved)
9	0	0	Yes	Correctly predicted denial
10	1	1	Yes	Correctly predicted approval
11	0	1	No	False Positive (predicted approval, was denied)
12	0	1	No	False Positive (predicted approval, was denied)
13	0	1	No	False Positive (predicted approval, was denied)
14	1	1	Yes	Correctly predicted approval
15	1	1	Yes	Correctly predicted approval

Table 1: Actual vs. predicted labels for first 15 test records (0 = Not Approved, 1 = Approved)

5. Model Evaluation

5.1. Classification Metrics

The controlled model (max_depth=4) was evaluated on the 76-record test set using four classification metrics:

Metric	Score	What It Means
Accuracy	61.84%	61.8% of all predictions (approved and not approved) were correct
Precision	0.6275	Of all predicted approvals, 62.8% were actually approved
Recall	0.7619	Of all actual approvals, 76.2% were correctly identified
F1-Score	0.6882	Harmonic mean of precision and recall — balanced performance measure

Table 2: Classification metrics for the controlled model (max_depth=4)

Precision is the most important metric for this problem. In an insurance context, a False Positive (approving a claim that should have been denied) translates into a direct financial loss, money paid out on an invalid claim and may expose the insurer to fraud and reserve mis-estimation. Recall matters too, because under-approving legitimate claims damages customer trust and triggers complaint volume, but the immediate cost of a paid-out invalid claim makes Precision the priority.

Metric trade-off: Precision is currently 0.628, which means almost 4 in every 10 model approvals are wrong. Recall is higher (0.762), so the model captures most legitimate approvals, but it does so by approving too liberally. In any production deployment, Precision should be treated as the floor constraint, a target of ≥ 0.80 should be reached before the model is used to support real approval decisions.

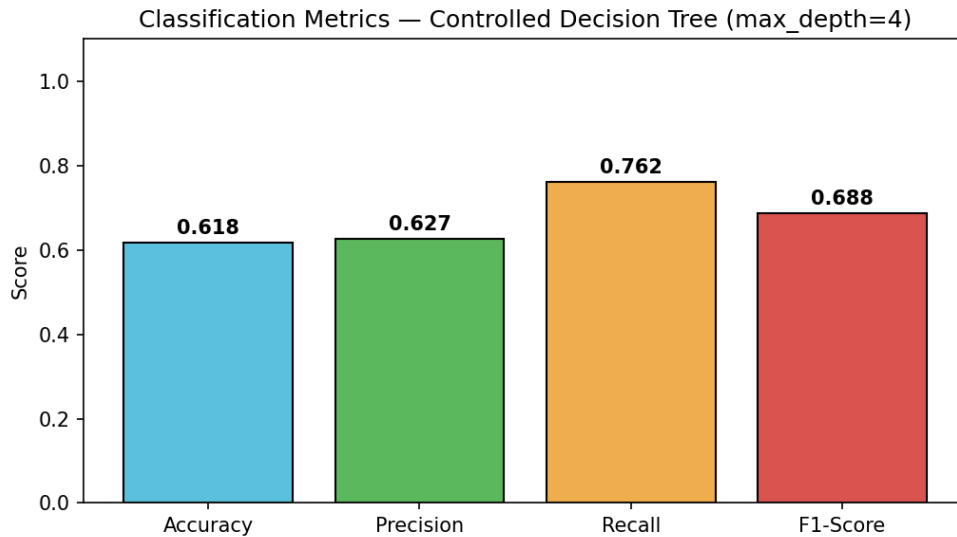


Figure 2: Classification metrics bar chart (Accuracy, Precision, Recall, F1-Score for the controlled model)

5.2. Confusion Matrix

The confusion matrix below shows the breakdown of all 76 test predictions:

	Predicted: Not Approved	Predicted: Approved
Actual: Not Approved	True Negative (TN): 15	False Positive (FP): 19
Actual: Approved	False Negative (FN): 10	True Positive (TP): 32

Table 3: Confusion matrix

- **True Negatives (15):** Model correctly denied claims that should have been denied (correct underwriting protection).
- **False Positives (19):** Model approved claims that should have been denied (direct payout losses, possible fraud exposure).
- **False Negatives (10):** Model denied claims that were valid (customer complaints, appeals workload, possible regulatory escalation).
- **True Positives (32):** Model correctly approved valid claims (core service delivery).

False Positives are the more serious error in an insurance context. Paying out money on a claim that should never have been approved represents a direct, often unrecoverable financial loss. False Negatives, while operationally damaging through complaints and churn, can typically be remediated through human appeals processes. The model produced 19 False Positives against 10 False Negatives, almost 2× more in the more financially damaging direction.

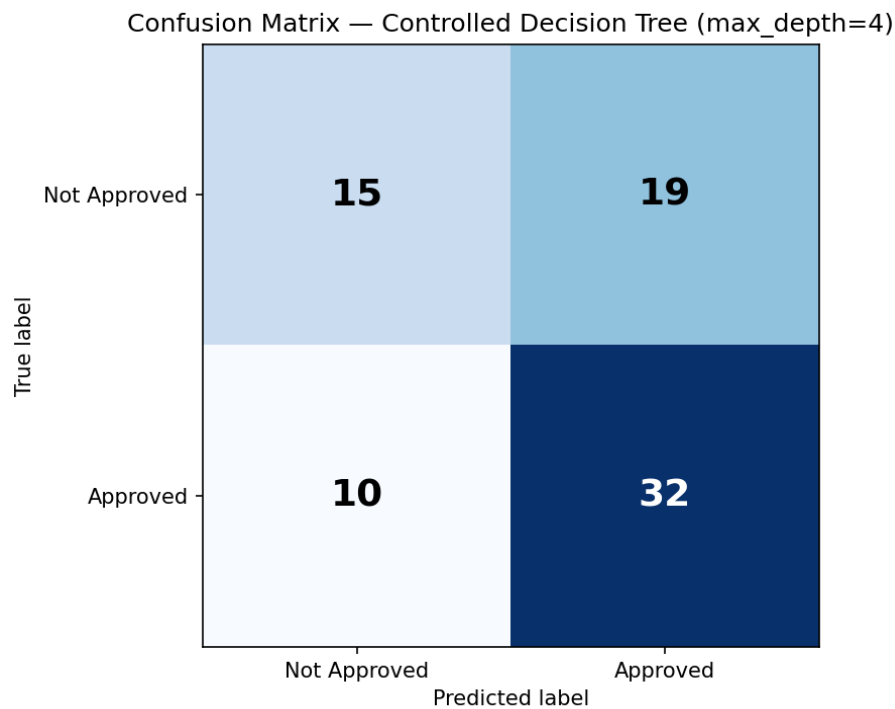


Figure 3: Confusion matrix heatmap

6. Error Analysis

Of 76 test records, the model made 29 wrong predictions: 19 False Positives and 10 False Negatives. False Positives are the most common error; the model more frequently approves claims that should be denied than the reverse.

The model systematically over-approves claims. This means the primary operational risk is payout leakage: paying out on claims that should have been declined. Scaled across a real claim's portfolio of thousands of submissions per quarter, this would translate into material financial losses and increased exposure to fraudulent submissions.

The error analysis reveals that the model leans in the operationally more damaging direction for insurance: over-approval is more harmful than under-approval, because paid-out claims are difficult to recover, while denied-but-valid claims can be caught through human appeals review. The 19 False Positives represent the riskiest predictions and should be the focus of any further model improvement.

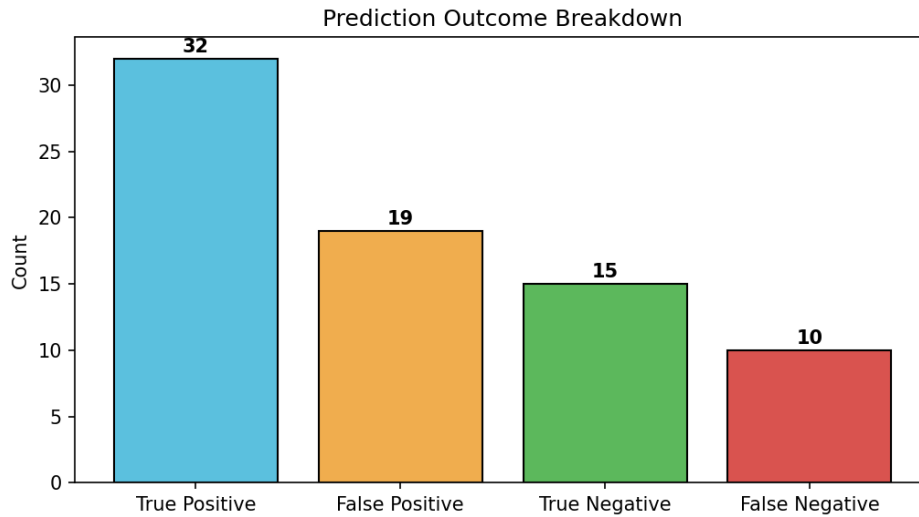


Figure 4: Error type counts

7. Cross-Validation

5-fold cross-validation was applied to the training set (304 records) to obtain a more reliable performance estimate than a single train/test split. In each fold, 80% of training data trains the model and 20% is used to validate it, rotating through all five folds.

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean Accuracy
52.46%	54.10%	57.38%	62.30%	56.67%	56.58% (Std: 0.034)

Table 4: 5-fold cross-validation results

The low standard deviation (0.034) confirms the model performs reasonably consistently across different subsets of the data; it is not sensitive to which records fall in the training vs. validation set.

However, the mean CV accuracy of 56.58% is meaningfully lower than the single-split test accuracy of 61.84%. This is a major concern. The 5-percentage-point gap suggests the single-split test result was slightly optimistic, the 76-record test set may have happened to be easier than average. The CV mean of 56.58% is the more conservative and reliable estimate of how the model would perform on genuinely new claims. At 56.58%, the model is only marginally above the 55.0% baseline of always predicting the majority class and must be treated as a starting point rather than a production-ready tool.

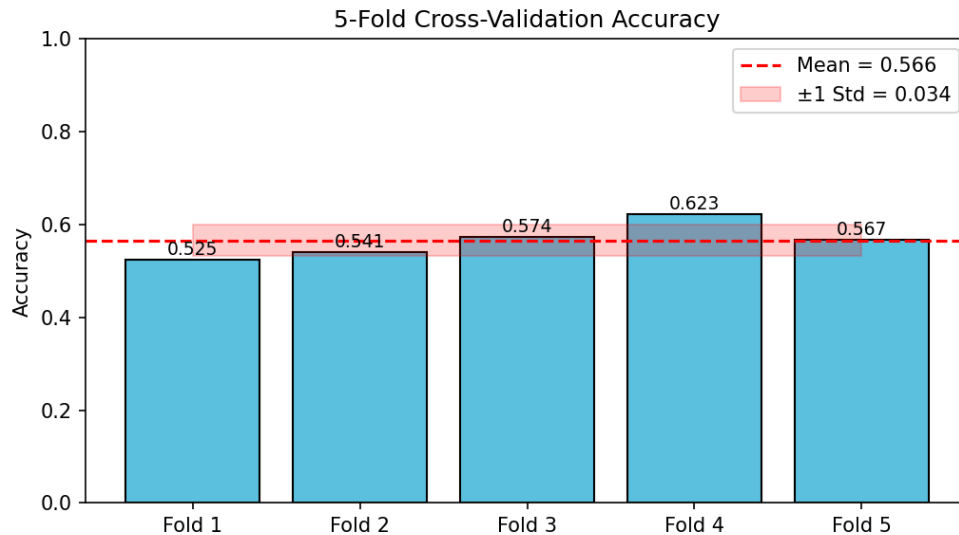


Figure 5: Cross-validation fold accuracy chart

8. Overfitting and Underfitting Analysis

The model was retrained across tree depths 1 through 15 to identify the optimal complexity level:

- **Depths 1–2:** Both training and testing accuracy are low, the model underfits, too simple to capture meaningful patterns.
- **Depths 3–4:** Both accuracies rise, the model learns genuine patterns.
- **Depths 5+:** Training accuracy continues to climb while test accuracy plateaus or falls, classic overfitting. The unconstrained tree (depth 10) achieves high training accuracy but lower test accuracy.

The controlled model at max_depth=4 was selected as the optimal balance, but the train/test gap remains substantial:

- **Training accuracy:** 78.62%
- **Testing accuracy:** 61.84%
- **Gap:** 16.78 percentage points, substantial, indicating residual overfitting even at the controlled depth.

All subsequent evaluations use the max_depth=4 model. Note that the cross-validation mean accuracy (56.58%) sitting below the test accuracy (61.84%) reinforces the picture: the model learns the training data more reliably than it generalises, and the realistic out-of-sample performance estimate is closer to 57% than the headline 62%.

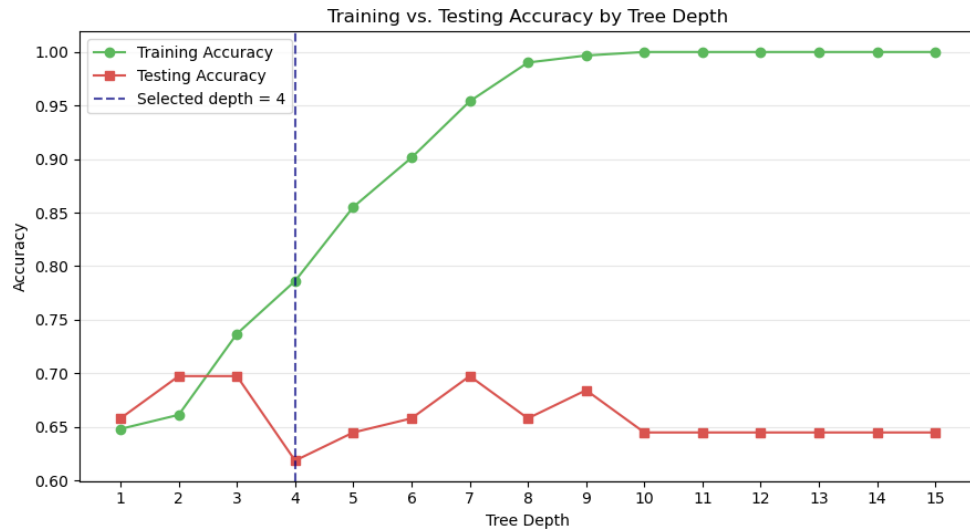


Figure 6: Training vs. testing accuracy by tree depth [overfitting visible at depth 5+; max_depth=4 selected as the optimal balance]

9. Feature Importance

Feature importances were extracted from the controlled tree (max_depth=4). The top features are:

Feature	Importance Score	Rank
payment_history_score	0.281	1
customer_tenure_years	0.166	2
police_report_filed_No	0.136	3
driving_experience_years	0.135	4
annual_premium	0.096	5
previous_claims	0.077	6
coverage_amount	0.075	7
claim_amount	0.034	8

Table 5: Top feature importance scores from the controlled tree (max_depth=4)

Business interpretation of top features:

- **payment_history_score (0.281):** The strongest predictor. Claimants with stronger payment track records are significantly more likely to be approved, this aligns with insurer trust signals: customers who pay reliably are treated as lower-risk and face less claim friction.
- **customer_tenure_years (0.166):** Length of relationship with the insurer is a powerful signal. Longer-tenured customers carry an established history with the insurer, making approval more likely. Retention is not only commercially valuable but also operationally meaningful.

- **police_report_filed_No (0.136):** Whether a police report was filed is one of the strongest single signals. The presence of a police report acts as third-party verification of the incident, materially shifting the approval decision.
- **driving_experience_years (0.135):** More experienced drivers are favoured, likely because experience correlates with lower-risk claims profiles and stronger underwriting trust.
- **annual_premium (0.096):** Premium level has moderate influence. Higher-premium policies typically carry broader coverage and stronger customer commitment, both of which correlate with approval.

Important caveat: feature importance scores show which features the Decision Tree used most, but they do not prove cause and effect. High importance for a feature means the model relied on it heavily, not that changing it will directly change approval outcomes. Claim amount itself sits surprisingly low (0.034), suggesting the model decides based on customer relationship and verification signals more than the size of the claim.

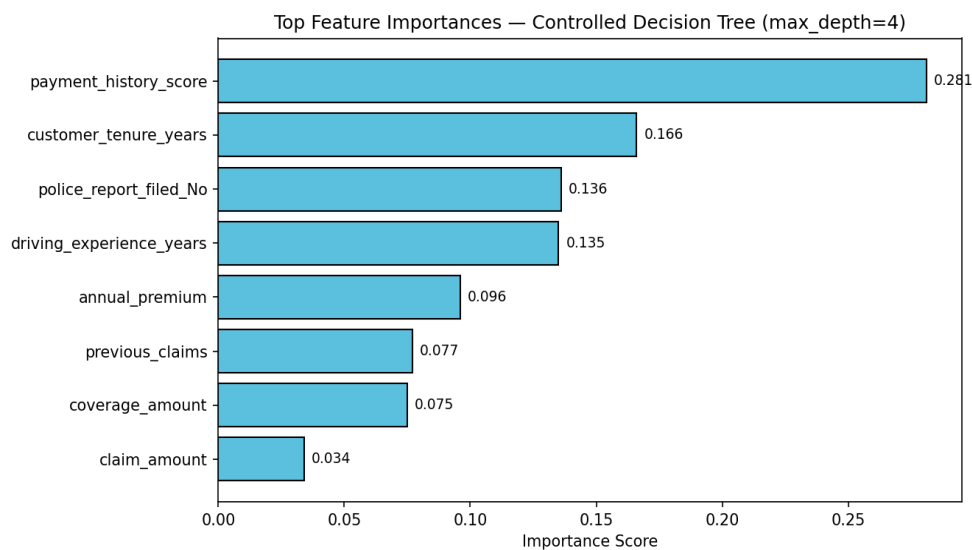


Figure 7: Top feature importance chart

10. SHAP Explanation

SHAP (SHapley Additive exPlanations) was used to explain the model beyond basic feature importance. Unlike global feature importances, SHAP provides both global patterns

and individual-level explanations, and crucially, it shows the direction of each feature's influence (increases or decreases the probability of approval).

The SHAP summary plot shows that `payment_history_score`, `customer_tenure_years`, and `police_report_filed` are the most consistently influential features across the entire test set. High values of these features (shown in red) push predictions strongly toward approval, while low values (blue) push toward denial.

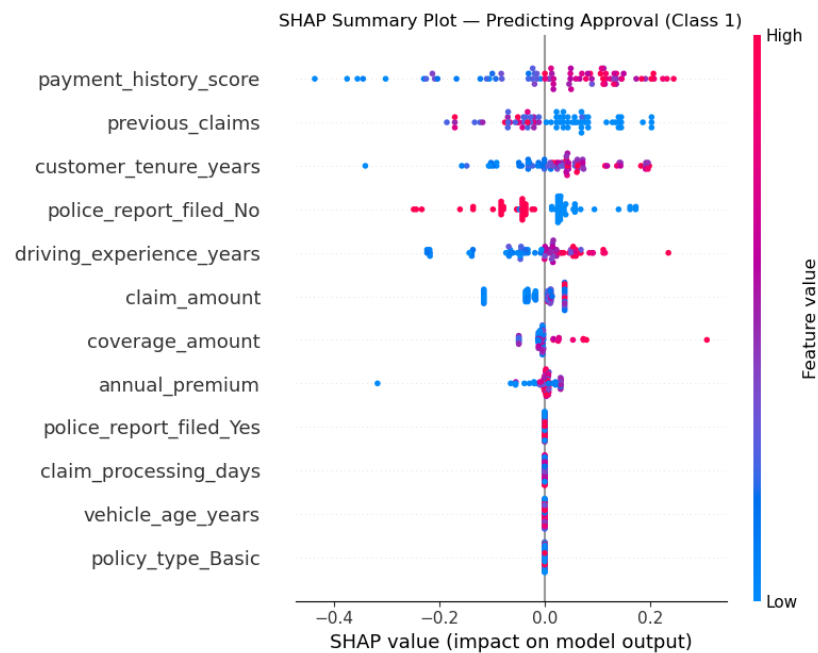


Figure 8: SHAP summary plot (class 1 — Approved) each row is a feature, each dot is a test record; red = high feature value, blue = low

For the individual record explained (Record 1, actual = Approved, predicted = Approved), SHAP reveals the specific features that pushed the prediction toward or away from approval:

Feature	SHAP Value (push toward approval)
customer_tenure_years	+0.137 (pushes toward approval)
police_report_filed_No	+0.057 (pushes toward approval)
annual_premium	+0.031 (pushes toward approval)
payment_history_score	-0.023 (pushes away from approval)
previous_claims	-0.021 (pushes away from approval)
claim_amount	-0.017 (pushes away from approval)
driving_experience_years	+0.016 (pushes toward approval)
coverage_amount	-0.004 (pushes away from approval)

Table 6: SHAP values for Record 1

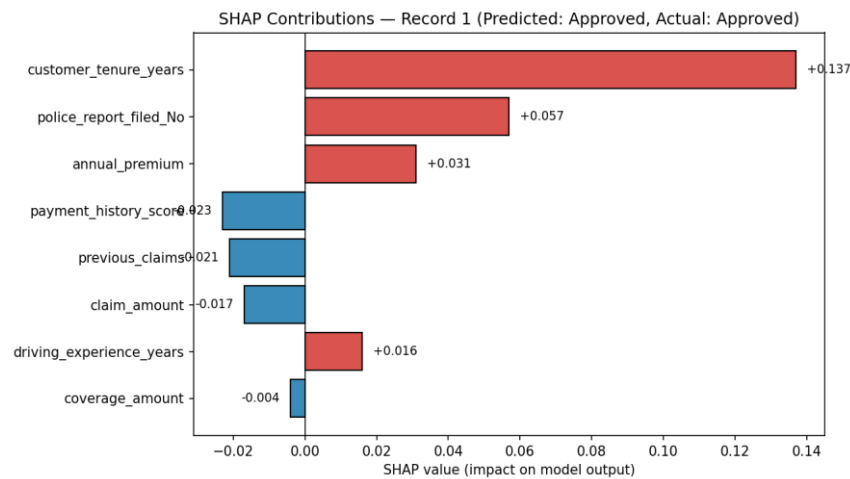


Figure 9: SHAP contribution chart for Record 1 decomposition of the individual prediction into per-feature contributions

SHAP goes beyond basic feature importance in two key ways:

- **Local explanations:** SHAP explains any individual claim decision, not just the global model behaviour.
- **Direction of influence:** Feature importance gives magnitude only. SHAP shows whether each feature increased or decreased the predicted probability of approval essential for auditable, defensible claim decisions in a regulated industry.

11. LIME Explanation

LIME (Local Interpretable Model-agnostic Explanations) was used to explain one individual prediction in plain language. For Record 1, LIME predicts a 72.5% probability of approval (27.5% probability of denial), resulting in a predicted class of Approved.

LIME works by creating a simplified local model around the specific record, testing small perturbations of the features, and identifying which features have the largest influence on the prediction in that local neighbourhood. The table below shows the feature weights for Record 1:

Feature Condition	Weight	Direction
police report filed No ≤ 0.00	+0.111	Supports Approval
previous claims > 1.00	-0.092	Against Approval
customer tenure years > 7.00	+0.091	Supports Approval
claim amount ≤ 3303.18	-0.065	Against Approval
coverage amount ≤ 25103.75	-0.022	Against Approval
$0.00 < \text{customer segment Loyal} \leq 1.00$	+0.016	Supports Approval

$59.00 < \text{payment_history_score} \leq 71.00$	-0.016	Against Approval
$20.00 < \text{driving_experience_years} \leq 35.00$	+0.012	Supports Approval
$\text{accident_severity_High} \leq 0.00$	+0.008	Supports Approval
$0.00 < \text{police_report_filed_Yes} \leq 1.00$	+0.004	Supports Approval

Table 7: LIME feature weights for Record 1 — positive weights support approval, negative weights work against it

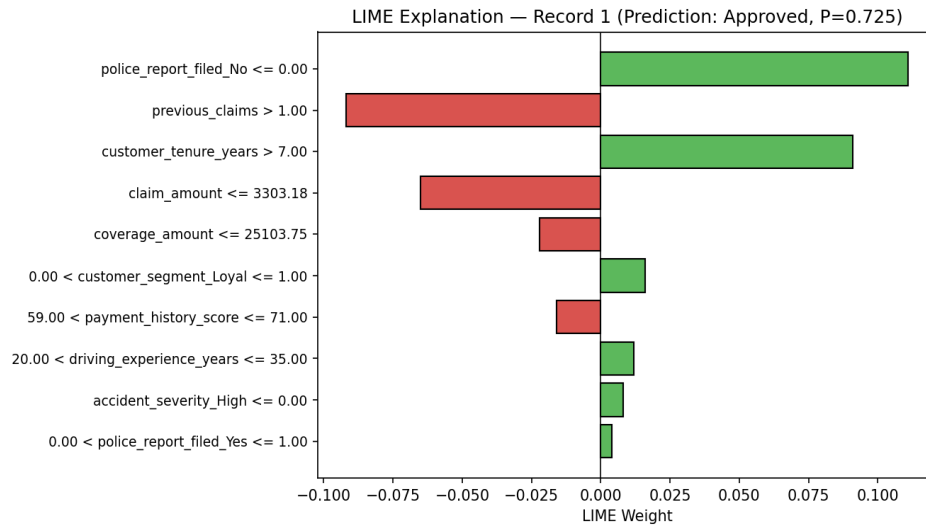


Figure 10: LIME feature weight chart for Record 1 — green bars support approval, red bars work against it

Features supporting the prediction of approval include the absence of a "no police report" condition (a third-party verification signal), long customer tenure of more than seven years, and being a Loyal customer segment. Features working against approval include a history of more than one previous claim and a relatively low claim amount combined with low coverage. The supporting factors outweighed the opposing ones, producing the 72.5% approval probability.

For business users: a claims adjuster reviewing a borderline approval can use the LIME explanation to identify exactly which conditions are driving the model's recommendation and decide whether the model's reasoning matches their own professional judgement. This makes individual model decisions auditable and defensible, which is essential in an insurance context where claimants, regulators, and internal auditors may all require explanations of automated decisions.

12. Fairness and Bias Reflection

Model performance was evaluated across three group-related columns: gender, region, and age group. These columns were excluded from model training but retained specifically for this analysis.

12.1. Gender

Group	N	Actual Positive Rate	Predicted Positive Rate	Accuracy
Female	31	45.2%	54.8%	58.1%
Male	45	62.2%	75.6%	64.4%

Table 8: Fairness metrics by gender

Accuracy is meaningfully lower for Female claimants (58.1%) than Male claimants (64.4%), a 6.3-point gap. More concerning, the predicted positive rate for Male claimants (75.6%) is 13.4 points higher than their actual approval rate (62.2%), indicating the model systematically over-approves Male claims. Female claimants are also over-approved (54.8% predicted vs 45.2% actual) but to a smaller degree. Although gender was excluded from training, the model has clearly learned it indirectly through correlated features, a form of proxy bias.

12.2. Region

Region	N	Actual Positive Rate	Predicted Positive Rate	Accuracy
East	19	63.2%	68.4%	63.2%
North	13	61.5%	76.9%	69.2%
South	17	52.9%	52.9%	64.7%
West	27	48.1%	70.4%	55.6%

Table 9: Fairness metrics by region

The West region has the lowest accuracy at 55.6%, the model is wrong on nearly one in every two West region claims. The model also significantly over-predicts approval for West region claimants (predicted 70.4% vs actual 48.1%), a 22-point over-approval gap. If claims teams rely on this model to triage incoming Western-region claims, the insurer would face systematic over-approval losses concentrated in that region. This disparity warrants investigation into whether the training data underrepresents West region claim patterns.

12.3. Age Group

Age Group	N	Actual Positive Rate	Predicted Positive Rate	Accuracy
18-25	12	41.7%	66.7%	41.7%
26-35	16	56.2%	43.8%	62.5%
36-50	21	66.7%	66.7%	71.4%
51+	27	51.9%	81.5%	63.0%

Table 10: Fairness metrics by age group

The age group breakdown reveals the most severe fairness disparities in the entire analysis. The 18–25 group shows just 41.7% accuracy, worse than random, combined with a 25-point over-approval gap (predicted 66.7% vs actual 41.7%). The 51+ group shows the largest over-approval gap of any segment in the analysis: the model over-predicts approval for this group by 29.6 percentage points (predicted 81.5% vs actual 51.9%). If the model were used to allocate adjuster review effort, both young and older claimants would be systematically over-approved without scrutiny.

12.4. Overall Fairness Assessment

The model does not perform equally across geographic, gender, or age groups. The dominant pattern is systematic over-approval concentrated in specific demographic and geographic segments, the financially riskiest direction for the insurer and the least fair direction for claimants. The West region, the 18–25 age group, and the 51+ age group show the weakest performance, both in absolute accuracy and in prediction-bias direction. These groups should not be subject to automated model decisions without human review.

Although gender, age group, and region were excluded from model training, the model has clearly learned them indirectly through correlated features (proxy bias). The model must be used strictly as a decision-support tool. Claims adjusters must retain final decision authority, especially for at-risk groups. Model predictions must never be used to deny claims or trigger fraud investigations for individual claimants based on model output alone. The specific fairness risk here is geographic and demographic over-approval, which can lead to inconsistent claim

outcomes in ways that correlate with protected characteristics — a material regulatory and reputational concern in insurance.

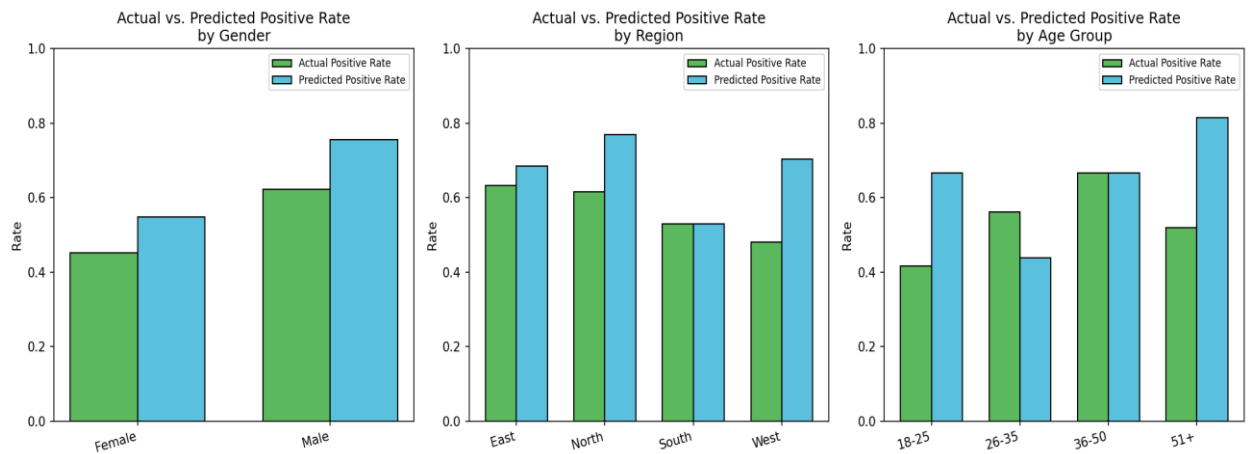


Figure 11: Actual vs. predicted positive rates by group (gender, region, age group)

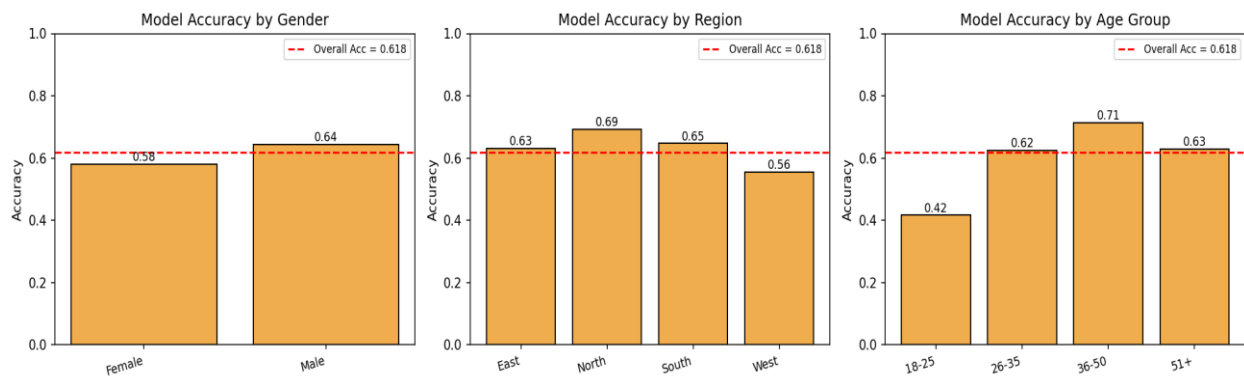


Figure 12: Model accuracy by group

13. Business Actions and Recommendations

Based on the model's findings, the following actions are recommended for the insurance claims team:

- Use the model strictly as a triage and decision-support tool. Claims adjusters should retain final approval authority on every claim, particularly any claim the model flags as high-confidence approval, since False Positives are the dominant error type.

- Prioritise reviewer time on the model's low-confidence and contested predictions. Adjusters should focus manual review on borderline scores rather than on cases the model is already confident about.
- Treat claims from the West region, 18–25 age group, and 51+ age group as requiring mandatory human review. The model is least reliable for these segments and systematically over-approves them; relying on automated approval here would expose the insurer to concentrated payout losses.
- Use SHAP and LIME outputs to generate explanation summaries for every appealed denial or contested approval. This creates an auditable trail for claimants, internal compliance, and regulators.
- Track Precision as the primary KPI for any future model iteration. The current 0.628 is too low for production deployment; a target of ≥ 0.80 should be reached before the model influences real approval decisions.
- Do not use model predictions as the sole basis for any individual-facing decision (e.g., denying a claim or triggering a fraud investigation). All model outputs should inform human decisions, not replace them.

14. Limitations

The following limitations should be considered before applying these results in a real claims operation:

- **Small dataset:** 380 records is insufficient for a robust production model. Results especially fairness metrics for sub-groups of 12–27 records may not generalise to the larger claims population.
- **CV vs. test gap:** The 5-fold CV mean accuracy of 56.58% is meaningfully lower than the single-split test accuracy of 61.84%. The CV estimate is more reliable and reveals the model is barely above the 55.0% naive baseline.
- **Train/test overfitting gap:** Even at the controlled depth of 4, the model overfits by 16.8 percentage points (78.6% training vs 61.8% testing accuracy).
- **Decision Tree limitations:** Decision Trees are prone to instability, small changes in training data can produce structurally different trees. Ensemble methods (Random Forest, Gradient Boosting) would likely produce more stable predictions.

- **Static model:** Claim approval patterns shift over time as fraud techniques evolve, policy terms change, and customer behaviour adjusts. The model requires periodic retraining.
- **Synthetic-style data:** The dataset was prepared for educational purposes. Real-world insurance data would include more complex patterns, missing data, claims fraud signals, and distribution shifts.
- **Binary target only:** The model predicts only approve/deny. It does not predict approval probability (which would enable risk-tiered triage) or partial approval (e.g., approving only part of a claimed amount).

15. Responsible AI Considerations

Insurance AI applications that affect claim outcomes and customer financial wellbeing carry a heightened responsibility to operate fairly, transparently, and within regulatory frameworks. The following principles should govern any deployment of this or similar models:

- **Fairness:** The model must never be used to disadvantage claimants from any geographic or demographic group. Model predictions must not trigger automatic denials, fraud flags, or reduced service quality for any subgroup.
- **Human oversight:** All predictions must be reviewed by a claim's professional before any approval or denial decision is finalised. The model serves as a risk-scoring tool, not a decision-maker.
- **Explainability:** SHAP and LIME explanations allow claims staff to understand why a specific claim was flagged. This supports accountability, helps staff identify unreliable predictions, and provides defensible justifications for appeals and regulatory inquiries.
- **Transparency to claimants:** Insurers using approval prediction models should inform claimants that their submission data is used in automated triage processes, in accordance with applicable privacy and consumer protection regulations.

- **No punitive use:** Model outputs should never be used to penalise individual claimants for example, by denying a claim or triggering a fraud investigation based on a low approval score alone.
- **Model monitoring:** If deployed, the model's fairness metrics should be re-evaluated quarterly, particularly for the West region, 18–25 age group, and 51+ age group, which showed the largest disparities in this analysis.

16. Conclusion

This assignment applied a Decision Tree classifier to an Insurance Claim Approval dataset to predict whether submitted claims would be approved. The full machine learning pipeline including preprocessing, model training, evaluation, explainability, and fairness analysis was implemented using scikit-learn with a Pipeline and ColumnTransformer to prevent data leakage.

The controlled model (`max_depth=4`) achieved a single-split test accuracy of 61.84% and an F1-score of 0.688. However, the 5-fold cross-validation mean accuracy of 56.58% only 1.6 percentage points above the naive 55.0% majority-class baseline reveals that the model has not yet learned reliably generalizable patterns. The CV score is the more honest estimate of generalisation performance, and it indicates very limited predictive power on genuinely new claims.

The most important predictive features are `payment_history_score`, `customer_tenure_years`, `police_report_filed`, and `driving_experience_years`. These are intuitive insurer-trust signals customers with strong payment histories, long relationships, and third-party-verified incidents are most likely to be approved. The claim amount itself plays a smaller role than might be expected, which is a useful insight: the model decides based on relationship and verification signals more than the size of the requested payout.

SHAP analysis confirms these features as the dominant positive drivers across the test set, while LIME enables individual-level explanations that allow adjusters to audit specific predictions. For Record 1, LIME revealed that long tenure (>7 years) and the presence of a police report supported approval, while a history of multiple previous claims pushed against it with the supporting factors winning out (72.5% approval probability). Together, these tools transform the model from a black box into an interpretable decision-support instrument.

The fairness analysis identified material disparities. The West region (55.6% accuracy with a 22-point over-approval gap), the 18–25 age group (41.7% accuracy worse than random with a 25-point over-approval gap), and the 51+ age group (63.0% accuracy with a 30-point over-approval gap) are all systematically over-approved by the current model. Combined with the dominant False Positive error pattern (19 vs 10 False Negatives), this means the model would expose the insurer to concentrated payout losses in specific demographic and geographic segments while simultaneously creating an unfair claims experience for those groups.

The recommended path forward is to treat this model as an early-stage baseline prototype: use it to identify the most relevant approval signals and to build intuition about decision patterns, but do not deploy it autonomously. Retraining on a larger, more representative dataset (target $\geq 1,500$ records) particularly with more West region and younger-age-group claims combined with stronger models (Random Forest, Gradient Boosting), formal fairness mitigation, and a mandatory human-in-the-loop review process, would be the highest-priority next steps before any production consideration. Precision should be tracked as the headline KPI, with a target of ≥ 0.80 before any deployment.